

1. INTRODUCTION

1.1 Introduction:

Human Resource Management System (HRMS) is a software application developed to automate and manage employee-related activities in an organization. The system helps the HR department maintain employee records, attendance, payroll, and leave management efficiently.

The main purpose of this project is to reduce manual work and improve accuracy in handling employee information. HRMS provides a centralized platform where administrators can manage all HR operations digitally.

The project is developed using Java, JSP, Servlets, JDBC, SQLite/MySQL database, HTML, CSS, and Bootstrap. The system provides secure login functionality and user-friendly interfaces for both administrators and employees.

The HRMS project helps organizations improve productivity, reduce paperwork, save time, and generate accurate reports.

1.2 Abstract:

The Human Resource Management System is a web-based application designed to simplify and automate HR operations. The system includes modules such as employee management, attendance management, payroll management, and leave management.

The application allows administrators to add, update, delete, and view employee records. It also helps in maintaining attendance details, generating salary information, and managing employee leave requests.

The project minimizes manual errors and improves efficiency through automation. The system stores all records in a database which provides security, consistency, and easy access to information.

The HRMS project is suitable for small and medium-scale organizations where employee data management is important.

1.3 Existing System and Need for System:

Existing System

In the traditional HR management process, organizations maintain employee records manually using registers, spreadsheets, and paper documents. This manual process creates several problems such as:

- Time-consuming record management
- Data redundancy
- Difficulty in searching employee information
- Payroll calculation errors
- Attendance management complexity
- Lack of security and backup
- Increased paperwork

Need for Proposed System

The proposed HRMS system is needed to:

- Automate HR operations
- Reduce manual work
- Improve data accuracy
- Generate reports quickly
- Provide secure employee record management
- Improve organizational efficiency
- Reduce administrative cost
- Maintain centralized employee database

1.4 Scope of System:

The scope of the Human Resource Management System includes:

Employee Management

- Add new employee details
- Update employee information
- Delete employee records
- Search employee details

Attendance Management

- Record employee attendance
- Track present and absent status
- Generate attendance reports

Payroll Management

- Generate employee salary details
- Manage salary records
- Calculate employee payroll

Leave Management

- Apply for leave
- Approve or reject leave requests
- Maintain leave history

Report Generation

- Employee reports
- Attendance reports
- Payroll reports
- Leave reports

1.5 Operating Environment – Hardware and Software:

Hardware Requirements:

Processor: Intel i3 or above

RAM: 4 GB minimum

Hard Disk: 250 GB minimum

Software Requirements:

Operating System: Windows 10 or above

Frontend: HTML, CSS, Bootstrap

Backend: Java, JSP, Servlet

Database: SQLite

Server: Apache Tomcat

2. PROPOSED SYSTEM

2.1 Proposed System:

The proposed Human Resource Management System (HRMS) is a computerized web-based application developed to automate and simplify various HR-related operations within an organization. The system is designed to efficiently manage employee records, attendance tracking, payroll processing, and leave management through a single integrated platform. It helps administrators maintain organizational data accurately while reducing manual work and paperwork.

The HRMS enables administrators to easily add, update, delete, and manage employee information, attendance records, salary details, and leave requests. Employees can also interact with the system to check attendance, apply for leave, and view their personal details. By automating these activities, the system saves time, improves accuracy, enhances data security, and provides quick access to information through a centralized database system.

The system includes several important features such as secure login authentication, employee information management, attendance management, payroll processing, leave request management, report generation, database management, and a user-friendly interface. These features help improve organizational efficiency and provide better employee management in a reliable and systematic manner.

2.2 Feasibility Study:

A feasibility study determines whether the proposed system is practical and beneficial.

1. Technical Feasibility: The project is technically feasible because the required technologies such as Java, JSP, Servlets, JDBC, SQLite/MySQL, and Apache Tomcat are easily available and supported.

2. Economic Feasibility: The system is cost-effective because it reduces manual labor and paperwork. The software tools used are open-source or freely available.

3. Operational Feasibility: The system is user-friendly and easy to operate. HR staff can easily learn and use the system.

4. Schedule Feasibility: The project can be completed within the specified time using proper planning and development methodology

2.3 Objectives of the System:

The major objectives of the HRMS project are:

- To automate HR operations
- To maintain employee records digitally
- To reduce manual errors
- To improve data security
- To simplify attendance tracking
- To generate payroll reports efficiently
- To improve leave management process
- To provide fast access to employee information

2.4 Users of the System:

Administrator

The administrator manages all HR operations including:

- Employee management
- Attendance management
- Payroll management
- Leave approval
- Report generation

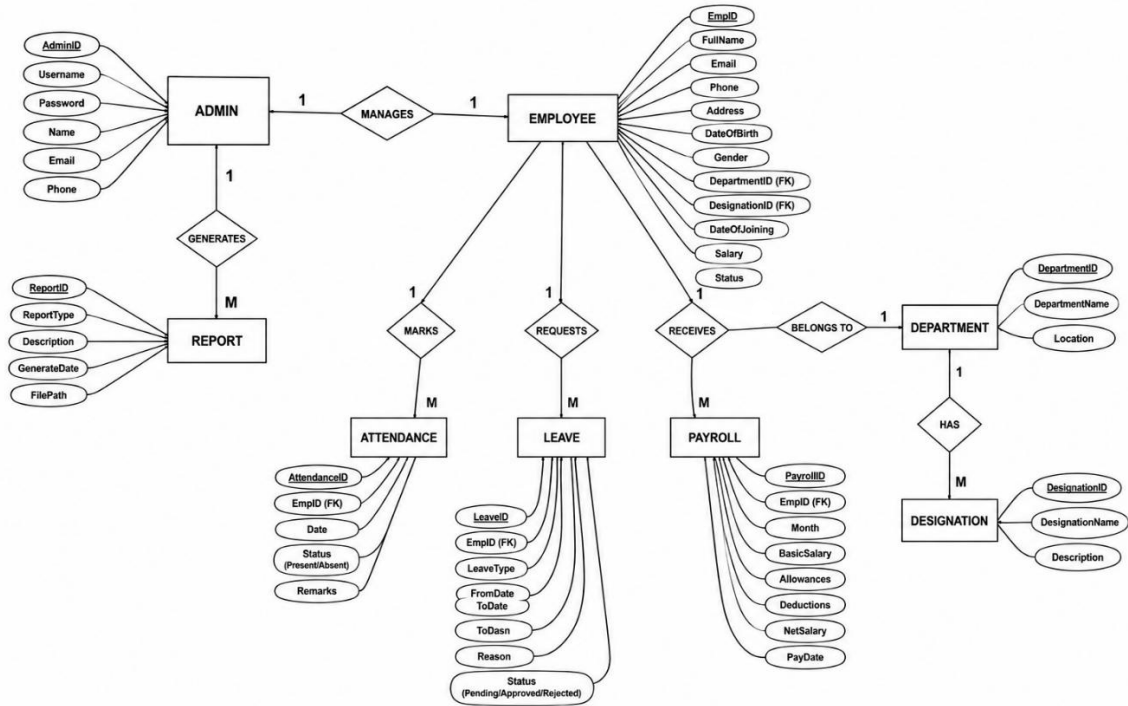
Employee

Employees can:

- View personal details
- Check attendance
- Apply for leave
- View salary details

3. ANALYSIS & DESIGN

3.1 Entity Relationship Diagram (ERD):



3.2 Table Structure:

Employee Table:

Field Name	Data Type	Description
employee_id	INT	Primary Key
name	VARCHAR(100)	Employee Name
email	VARCHAR(100)	Employee Email
department	VARCHAR(50)	Department Name
designation	VARCHAR(50)	Job Position
salary	DOUBLE	Employee Salary

Attendance Table:

Field Name	Data Type	Description
attendance_id	INT	Primary Key
employee_id	INT	Foreign Key
attendance_date	DATE	Attendance Date
status	VARCHAR(20)	Present/Absent

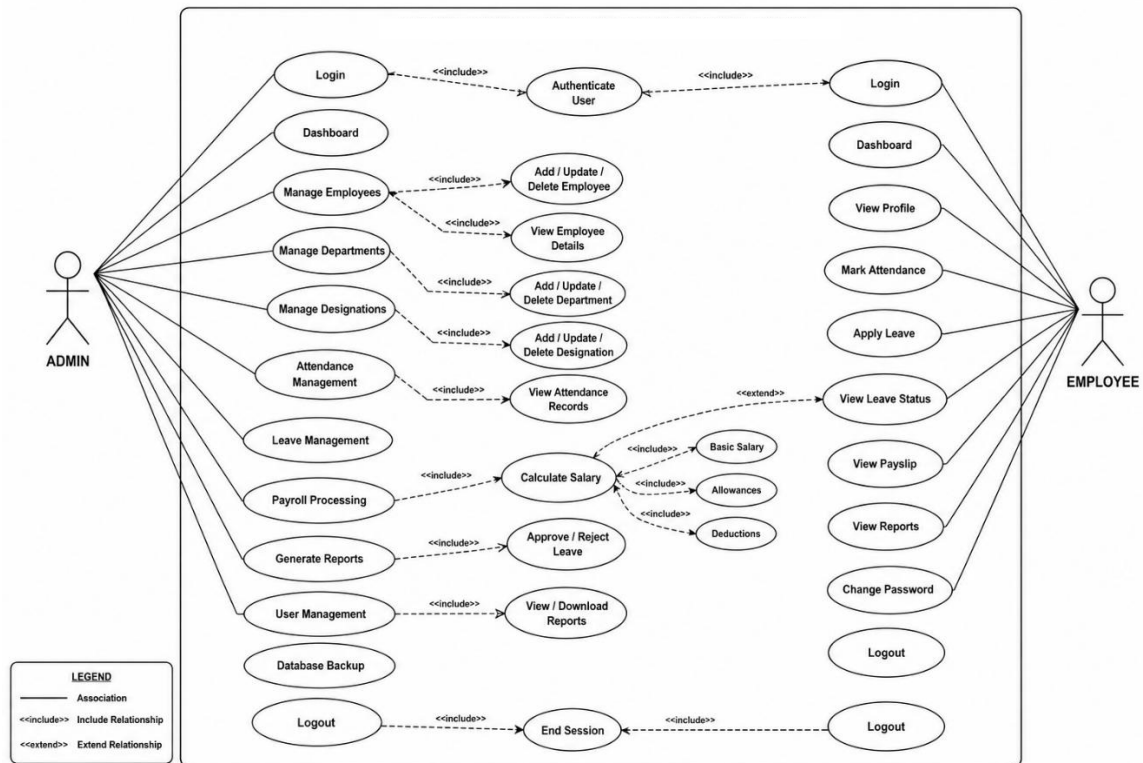
Leave Table:

Field Name	Data Type	Description
leave_id	INT	Primary Key
employee_id	INT	Foreign Key
leave_type	VARCHAR(50)	Leave Category
start_date	DATE	Leave Start Date
end_date	DATE	Leave End Date
status	VARCHAR(20)	Approved/Rejected

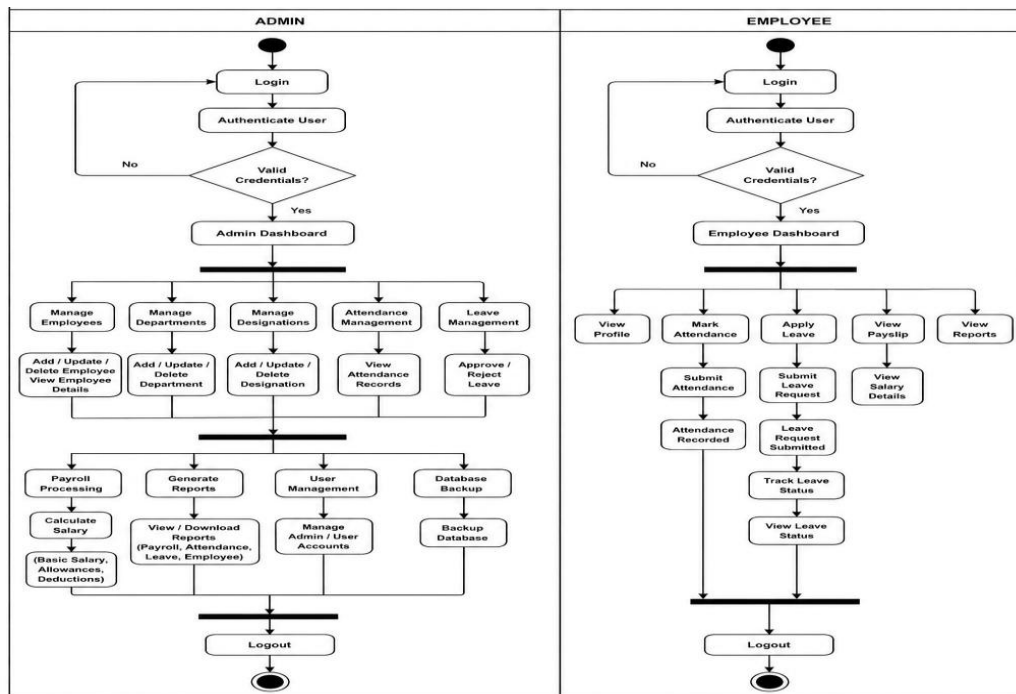
Payroll Table:

Field Name	Data Type	Description
payroll_id	INT	Primary Key
employee_id	INT	Foreign Key
basic_salary	DOUBLE	Basic Salary
bonus	DOUBLE	Bonus Amount
deduction	DOUBLE	Deduction Amount
net_salary	DOUBLE	Final Salary

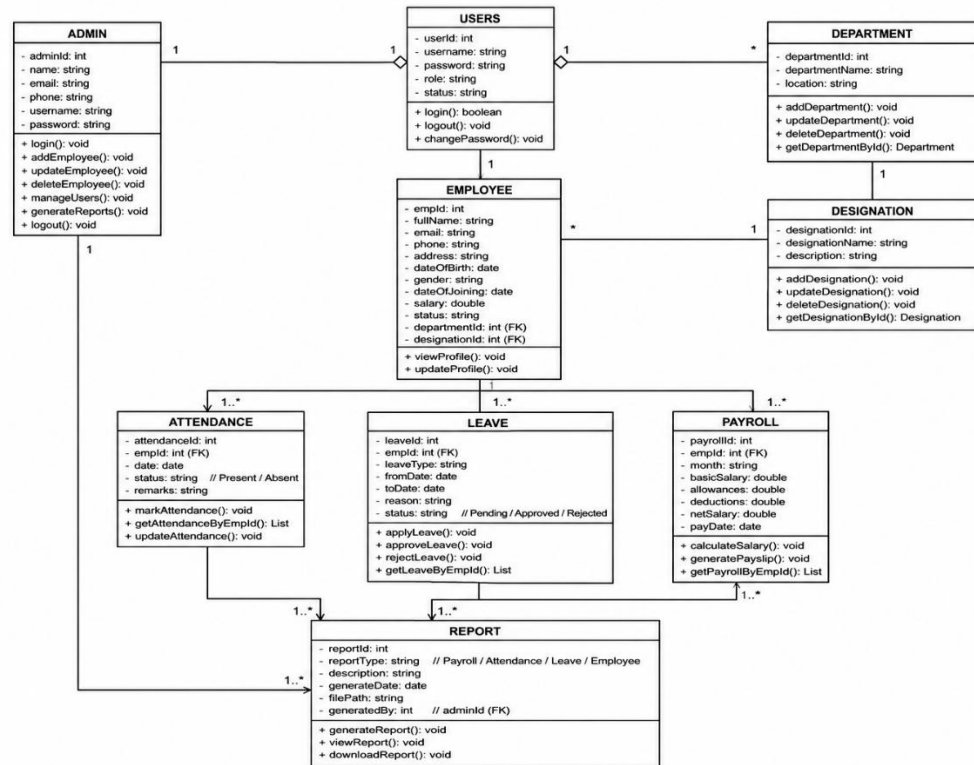
3.3 Use Case Diagram:



3.4 Activity Diagram:



3.5 Class Diagram:



3.6 Simple Input and Output Screens:

Registration:

HR

Join HRMS

Create your account to get started

Full Name

Email Address

Password

Department

Engineering
▼

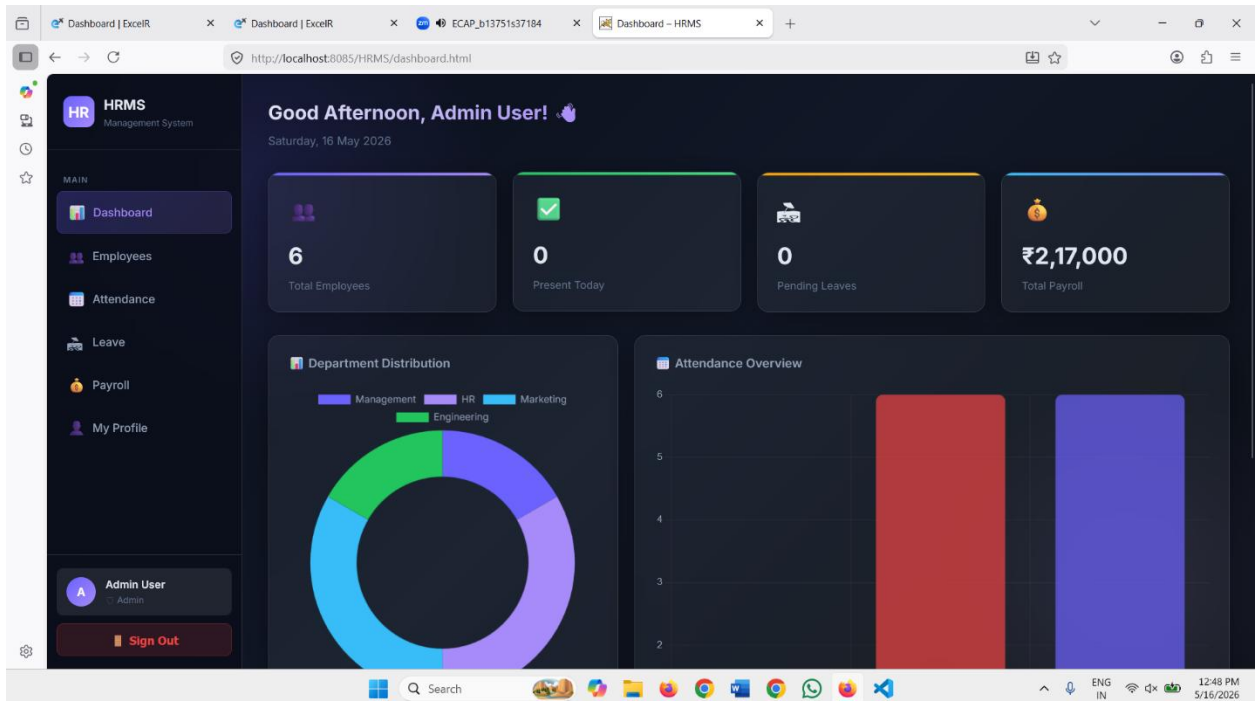
Create Account

Already have an account? [Sign In](#)

Login:

The screenshot shows a web browser window with the URL `http://localhost:8085/HRMS/login.html`. The page features a dark blue background with a central white login form. The form includes a purple 'HR' logo, the title 'HR Management System', and the instruction 'Sign in to continue to your dashboard'. It has two input fields: 'Email Address' with the value 'admin@hrms.com' and 'Password' with masked characters. A purple 'Sign In' button is at the bottom, along with a link to 'Register here' for new users.

Admin Login Dashboard:



Employees:

The screenshot shows the 'Employee Management' page of the HRMS. The left sidebar contains navigation links: Dashboard, Employees (selected), Attendance, Leave, Payroll, and My Profile. The main content area has a search bar and a dropdown for 'All Departments'. Below this is a table of employees with columns for ID, Name, Email, Department, Role, Salary, and Actions (Edit, Delete).

#	NAME	EMAIL	DEPARTMENT	ROLE	SALARY	ACTIONS
1	Nikita Ambre	nikitaambre1@gmail.com	Management	employee	₹60,000	Edit Delete
2	Shital Khilari	shital1@gmail.com	HR	employee	₹60,000	Edit Delete
3	ankita desai	ankita1@gmail.com	HR	employee	₹0	Edit Delete
4	rohit sharma	rohit1@gmail.com	Marketing	employee	₹57,000	Edit Delete
5	shubham dighe	shubhamdighe@prens.in	Marketing	employee	₹0	Edit Delete
6	virat kohali	virat1@gmail.com	Engineering	employee	₹40,000	Edit Delete

The bottom of the screen shows a Windows taskbar with the date and time: 12:48 PM 5/16/2026.

Attendance:

The screenshot shows the 'Attendance' page of the HRMS. The left sidebar is the same as the previous page, with 'Attendance' selected. The main content area has a 'Mark Employee Attendance' section with a 'Select Employee' dropdown and 'Mark Present'/'Mark Absent' buttons. Below this is a table of 'All Employee Attendance Records' with columns for ID, Employee, Date, and Status.

#	EMPLOYEE	DATE	STATUS
1	Nikita Ambre	09/05/2026	Absent
2	Shital Khilari	09/05/2026	Present
3	shubham dighe	09/05/2026	Present

The bottom of the screen shows a Windows taskbar with the date and time: 12:49 PM 5/16/2026.

Leave:

The screenshot displays the 'Leave Management' section of the HRMS. The left sidebar contains navigation links: Dashboard, Employees, Attendance, Leave (selected), Payroll, and My Profile. The main content area is titled 'Leave Management' with the subtitle 'Manage and track leave requests'. It features a 'Leave Requests' table with a search filter and the following data:

#	EMPLOYEE	REASON	FROM	TO	STATUS	ACTIONS
1	Nikita Ambre	medical	08/05/2026	08/05/2026	Approved	—

The bottom of the interface shows the user 'Admin User' and a 'Sign Out' button. The Windows taskbar at the bottom indicates the time is 12:49 PM on 5/16/2026.

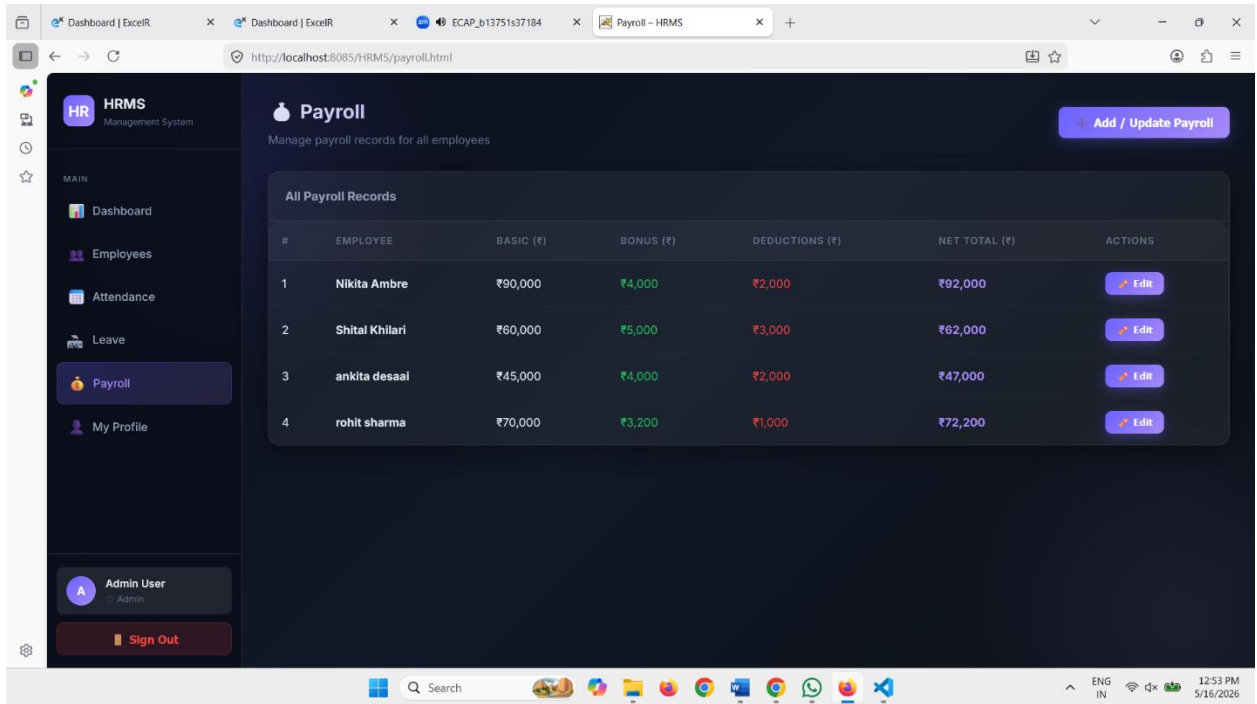
Add/ Update Payroll:

The screenshot shows the 'Payroll' section of the HRMS with the 'Set Payroll' modal open. The modal contains the following fields and values:

- Employee: Shital Khilari (HR)
- Basic Salary (₹): 60000
- Bonus (₹): 5000
- Deductions (₹): 3000
- Net Total (₹) (auto): ₹62,000

The modal includes 'Cancel' and 'Save Payroll' buttons. The background shows the 'Payroll' section of the HRMS interface. The Windows taskbar at the bottom shows the time is 12:51 PM on 5/16/2026.

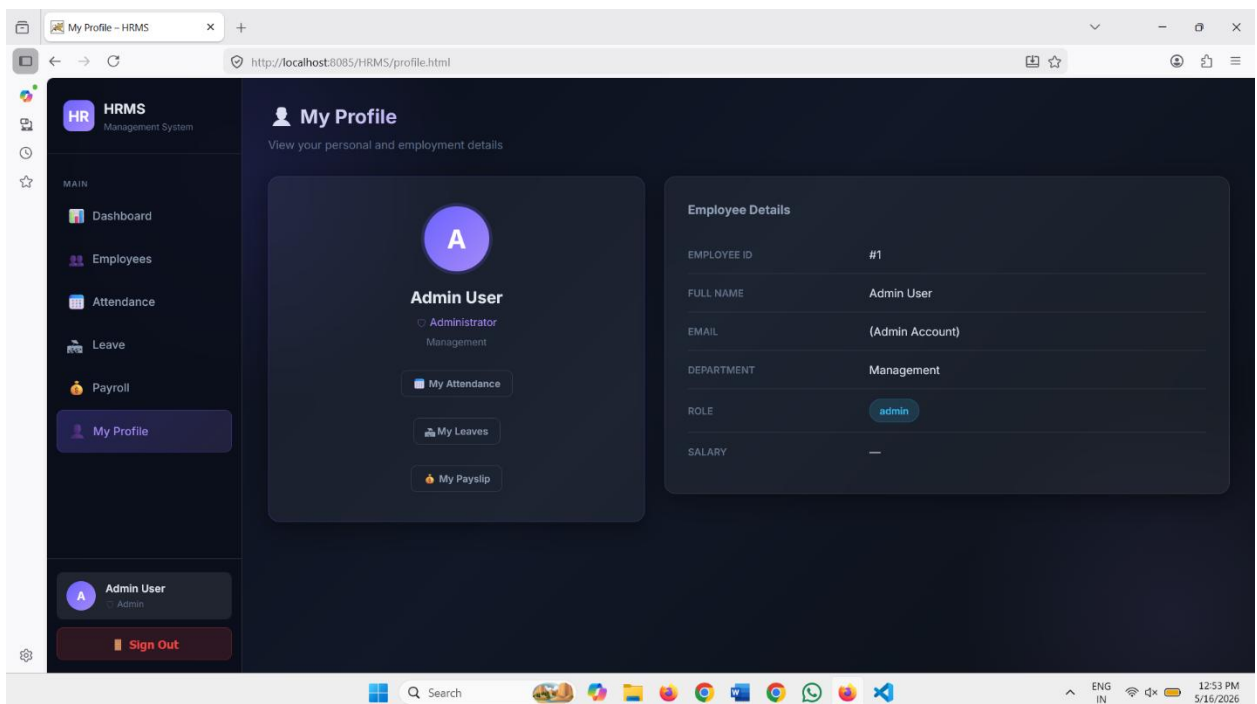
Payroll:



The screenshot displays the 'Payroll' section of the HRMS interface. The left sidebar contains navigation links: Dashboard, Employees, Attendance, Leave, Payroll (selected), and My Profile. The main content area is titled 'Payroll' with a subtitle 'Manage payroll records for all employees' and an 'Add / Update Payroll' button. Below this is a table titled 'All Payroll Records' with columns: #, EMPLOYEE, BASIC (₹), BONUS (₹), DEDUCTIONS (₹), NET TOTAL (₹), and ACTIONS. The table lists four employees with their respective financial details. The bottom of the interface shows the user 'Admin User' and a 'Sign Out' button.

#	EMPLOYEE	BASIC (₹)	BONUS (₹)	DEDUCTIONS (₹)	NET TOTAL (₹)	ACTIONS
1	Nikita Ambre	₹90,000	₹4,000	₹2,000	₹92,000	Edit
2	Shital Khilari	₹60,000	₹5,000	₹3,000	₹62,000	Edit
3	ankita desai	₹45,000	₹4,000	₹2,000	₹47,000	Edit
4	rohit sharma	₹70,000	₹3,200	₹1,000	₹72,200	Edit

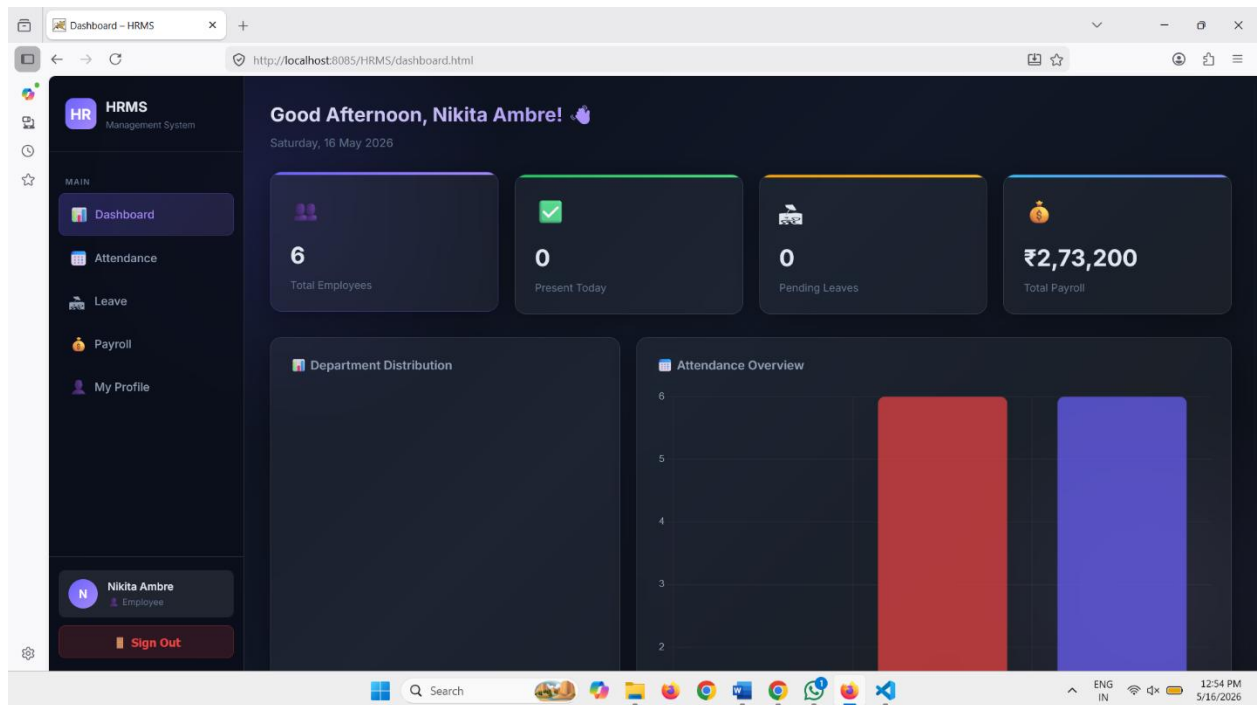
Admin Profile:



The screenshot shows the 'My Profile' section of the HRMS interface. The left sidebar is identical to the previous screenshot, with 'My Profile' selected. The main content area is titled 'My Profile' with a subtitle 'View your personal and employment details'. It features a central profile card for 'Admin User' (Administrator, Management) with links for 'My Attendance', 'My Leaves', and 'My Payslip'. To the right is an 'Employee Details' table. The bottom of the interface shows the user 'Admin User' and a 'Sign Out' button.

Employee Details	
EMPLOYEE ID	#1
FULL NAME	Admin User
EMAIL	(Admin Account)
DEPARTMENT	Management
ROLE	admin
SALARY	—

Employee Login Dashboard:



Attendance:

The screenshot shows the HRMS Attendance page for Nikita Ambre. The page is titled "Attendance" and "Today: Saturday, 16 May 2026". It features a sidebar with navigation links: Dashboard, Attendance, Leave, Payroll, and My Profile. The main content area displays a table titled "My Attendance Records" with a "Filter..." button. The table has columns for #, DATE, and STATUS. The records show three entries: 1 (16/05/2026, Present), 2 (09/05/2026, Absent), and 3 (08/05/2026, Present). The user profile at the bottom left shows "Nikita Ambre" as an Employee with a "Sign Out" button.

#	DATE	STATUS
1	16/05/2026	Present
2	09/05/2026	Absent
3	08/05/2026	Present

Apply for Leave:

The screenshot shows a web browser window with the URL `http://localhost:8085/HRMS/leave.html`. The page displays a modal titled "Apply for Leave" with the following fields:

- Reason ***: A text input field containing the word "festival".
- From Date ***: A date picker showing "10 / 13 / 2026".
- To Date ***: A date picker showing "10 / 16 / 2026".

At the bottom of the modal are two buttons: "Cancel" and "Submit Application".

Leave:

The screenshot shows the "Leave Management" page in the HRMS system. The page title is "Leave Management" with the subtitle "Apply for leave and track your requests". A sidebar on the left contains navigation links: "Dashboard", "Attendance", "Leave" (highlighted), "Payroll", and "My Profile". The main content area displays a table of "Leave Requests" with a "Filter..." button. The table has columns for "#", "REASON", "FROM", "TO", and "STATUS".

#	REASON	FROM	TO	STATUS
1	festival	13/10/2026	16/10/2026	Pending
2	medical	08/05/2026	08/05/2026	Approved

At the bottom of the sidebar, the user's profile is shown: "Nikita Ambre" (Employee), with a "Sign Out" button.

Payroll:

The screenshot shows the Payroll page of the HRMS system. The left sidebar contains a navigation menu with options: Dashboard, Attendance, Leave, Payroll (selected), and My Profile. The main content area is titled 'Payroll' and 'View and manage payroll records'. It displays a payroll summary for Nikita Ambre, showing a Basic Salary of ₹90,000, a Bonus of +₹4,000, Deductions of -₹2,000, and a Net Take-Home of ₹92,000. The bottom of the page shows a Windows taskbar with various application icons and a system clock indicating 12:56 PM on 5/16/2026.

Category	Amount
Basic Salary	₹90,000
Bonus	+₹4,000
Deductions	-₹2,000
Net Take-Home	₹92,000

Employee Profile:

The screenshot shows the My Profile page of the HRMS system. The left sidebar contains a navigation menu with options: Dashboard, Attendance, Leave, Payroll, and My Profile (selected). The main content area is titled 'My Profile' and 'View your personal and employment details'. It displays a profile card for Nikita Ambre, showing a profile picture, name, and role (Employee Management). Below the profile card are buttons for 'My Attendance', 'My Leaves', and 'My Payslip'. To the right, there is a table of Employee Details.

Employee Details	
EMPLOYEE ID	#17
FULL NAME	Nikita Ambre
EMAIL	nikitaambre1@gmail.com
DEPARTMENT	Management
ROLE	employee
SALARY	₹60,000

4. CODING

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Login - HRMS</title>

  <meta name="description" content="HR Management System - Secure Login Portal" />

  <!-- Main CSS -->

  <link rel="stylesheet" href="css/style.css" />

</head>

<body>

  <!-- Background -->

  <div class="gradient-bg"></div>

  <!-- Main Login Section -->

  <main class="login-page">

    <div class="glass-card login-card">

      <!-- Logo Section -->

      <div class="login-logo">

        <div class="logo-icon">

          HR

        </div>

        <h1>HR Management System</h1>

      </div>

    </div>

  </main>

</body>

</html>
```

```

    <p> Sign in to continue to your dashboard </p> </div>

<!-- Login Form -->

<form id="login-form" autocomplete="off" novalidate>

<!-- Email -->

    <div class="form-group"> <label for="email" class="form-label">

        Email Address

    </label>

    <input

        type="email"

        id="email"

        class="form-control"

        placeholder="e.g. admin@hrms.com"

        required

        autocomplete="off"

    /> </div>

<!-- Password -->

    <div class="form-group">

        <label for="password" class="form-label">

            Password

        </label>

        <input

            type="password"

            id="password"

```

```
        class="form-control"
        placeholder="Enter your password"
        required
        autocomplete="new-password" /> </div>
```

```
<!-- Error Message -->
```

```
<div
    id="login-error"
    style="
        color: var(--danger);
        font-size: 13px;
        margin-bottom: 12px;
        display: none;"></div>
```

```
<!-- Login Button -->
```

```
<button
    type="submit"
    id="login-btn"
    class="btn btn-primary login-btn">
    <span id="btn-text">
        Sign In
    </span>
</button>
```

```
</form>
```

```
<!-- Register Link -->
```

```

    <div class="login-hint">

Don't have an account?

<a href="register.html">

    <strong>Register here</strong> </a> </div> </div> </main>

<!-- JavaScript -->

    <script src="js/main.js"></script>

<script>

// Redirect user if already logged in

    if (getSession()) {

        window.location.href = 'dashboard.html';

    }

// Login Form Submit

document

    .getElementById('login-form')

    .addEventListener('submit', async (e) => { e.preventDefault();

// Get Input Values

    const email = document

        .getElementById('email')

        .value

        .trim();

    const password = document

        .getElementById('password')

        .value

```

```

        .trim();

// Elements

        const errEl = document.getElementById('login-error');

        const btnText = document.getElementById('btn-text');

// Reset UI

        errEl.style.display = 'none';

        btnText.textContent = 'Signing in...';

try {

// API Request

        const data = await apiPost('api/login', {

            email,

            password

        });

// Success

        if (data && data.success) {

            setSession({

                empId: data.empId,

                name: data.name,

                role: data.role

            });

            showToast(

                `Welcome back, ${data.name}!

                'success' );

```

```
        setTimeout(() => {  
            window.location.href = 'dashboard.html';  
        });  
    // Invalid Login  
    else {  
        errEl.textContent =  
            (data && data.message) ||  
            'Invalid credentials.';  
        errEl.style.display = 'block';  
    }  
    // Server Error  
    catch (err) {  
        errEl.textContent =  
            'Server error. Is Tomcat running?';  
        errEl.style.display = 'block';  
        btnText.textContent = 'Sign In'; } }  
</script>  
</body>  
</html>
```

5. DRAWBACKS AND LIMITATIONS

Drawbacks and Limitations

- Limited security features.
- No mobile app support.
- Requires internet connection.
- Basic report generation.
- Suitable for small organizations only.
- Limited user access control.

6. PROPOSED ENHANCEMENT

- Add mobile application support.
- Implement advanced security features.
- Add biometric attendance system.
- Enable email notifications.
- Improve report generation.
- Add cloud database support.
- Introduce employee self-service portal.
- Add real-time dashboard and analytics.

7. CONCLUSION

The Human Resource Management System (HRMS) successfully automates important HR activities such as employee management, attendance tracking, payroll processing, and leave management.

The system reduces manual work, saves time, improves accuracy, and helps in maintaining employee records efficiently.

It provides a user-friendly interface and centralized database system for better management of organizational data.

Overall, the project improves the efficiency and productivity of HR operations.

8. BIBLIOGRAPHY

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